RIDGE REGRESSION (L2)

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ALGORITHM EXPLANATION

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What is the Ridge Regression?

How does the algorithm work?

When should this algorithm be used?

Advantages and limitations

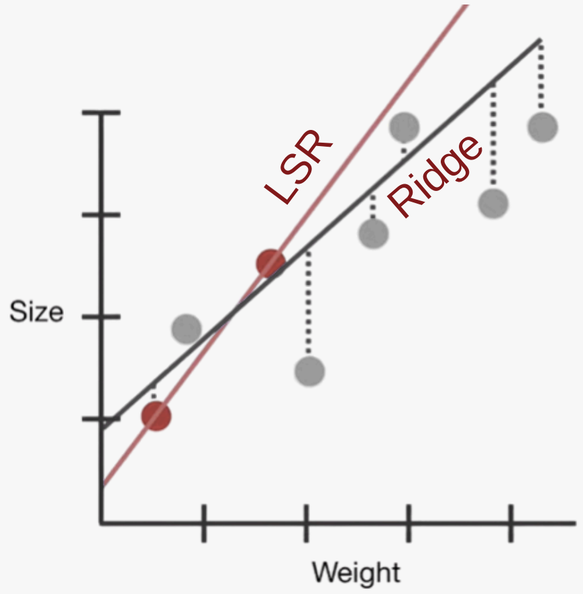
WHAT IS RIDGE REGRESSION (L 2)?

Ridge Regression is a machine learning algorithm used for regression problems.

$$y = \text{intercept} + \text{slope} \times \text{Weight}$$

Linear Regression + Model coefficients penalty = control size

HOW DOES IT WORK ?



1. RESIDUAL: THE ACTUAL VALUE - THE PREDICTED VALUE
2. LSR: MIN (SUM OF SQUARED RESIDUALS)
3. LSR: MIN (SUM OF SQUARED RESIDUALS + LAMBDA * SLOP^2)

WHEN SHOULD THIS RIDGE REGRESSION (L 2) BE USED?

1

High-Dimensional Data

Performs well when the number of features is large or exceeds observations.

2

All Features Matter

Ideal when most features contribute to the outcome and shouldn't be excluded (unlike L1, which zeros them out).

3

Handling Multicollinearity

It stabilizes the model when features are highly correlated by distributing the weights evenly among them instead of over-relying on one.

ADVANTAGES AND LIMITATIONS

Advantages

- Controlling overfitting
- Stability in Correlation
- Reduces variance and improves prediction on noisy datasets

limitations

- Does not eliminate irrelevant features
- Feature Scaling is Mandatory.
- Outliers Sensitivity

Exploratory Data Analysis (EDA)

Data Preprocessing

Model Implementation-

CODE IMPLEMENTATION

CODE IMPLEMENTATION

EDA

Data Preprocessing

Model Implementation-Evaluation

THANK YOU

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